

11.1

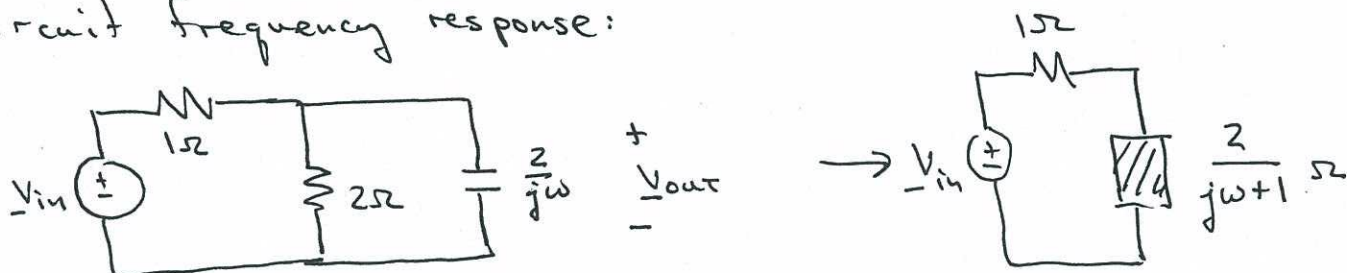
Input voltage: $V_{in}(t) = 3 \cos(2t - 20^\circ) - 2 \cos(3t + 30^\circ) + \cos(4t)$

$$\omega = 2 \Rightarrow \underline{V}_{in} = 3 \angle -20^\circ$$

$$\omega = 3 \Rightarrow \underline{V}_{in} = 2 \angle 30^\circ$$

$$\omega = 4 \Rightarrow \underline{V}_{in} = 1 \angle 0^\circ$$

Circuit frequency response:



$$\underline{V}_{out} = \underline{V}_{in} \left[\frac{\frac{2}{j\omega + 1}}{1 + \frac{2}{j\omega + 1}} \right] = \underline{V}_{in} \left[\frac{2}{j\omega + 3} \right]$$

$$\omega = 2 \Rightarrow \frac{2}{j\omega + 3} = 0.555 \angle -33.7^\circ$$

$$\omega = 3 \Rightarrow \frac{2}{j\omega + 3} = 0.471 \angle -45^\circ$$

$$\omega = 4 \Rightarrow \frac{2}{j\omega + 3} = 0.40 \angle -53.1^\circ$$

$$\text{Output: } \underline{V}_{out} = \underline{V}_{in} \left[\frac{2}{j\omega + 3} \right]$$

$$\omega = 2 \Rightarrow \underline{V}_{out} = (3 \angle -20^\circ)(0.555 \angle -33.7^\circ) = 1.66 \angle -53.7^\circ$$

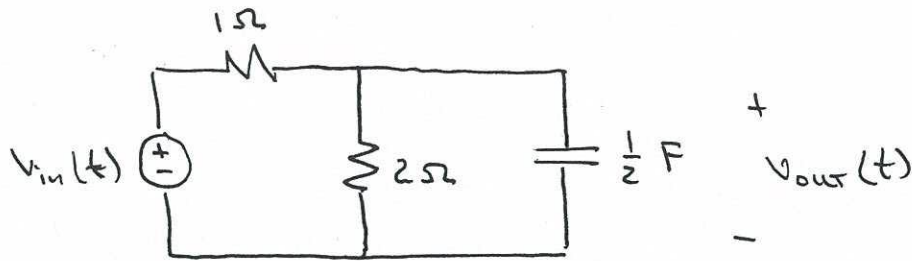
$$\omega = 3 \Rightarrow \underline{V}_{out} = (2 \angle 30^\circ)(0.471 \angle -45^\circ) = 0.94 \angle -15^\circ$$

$$\omega = 4 \Rightarrow \underline{V}_{out} = (1 \angle 0^\circ)(0.4 \angle -53.1^\circ) = 0.4 \angle -53.1^\circ$$

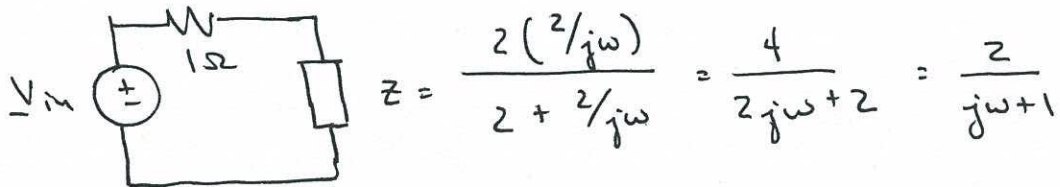
$$V_{out}(t) = 1.66 \cos(2t - 53.7^\circ) - 0.94 \cos(3t - 15^\circ) + 0.4 \cos(4t - 53.1^\circ)$$

$$i_{n(1)} - V_{out}(t) = 1.66 \cos(2t - 53.7^\circ) - 0.94 \cos(3t - 15^\circ) + 0.4 \cos(4t - 53.1^\circ)$$

11.2



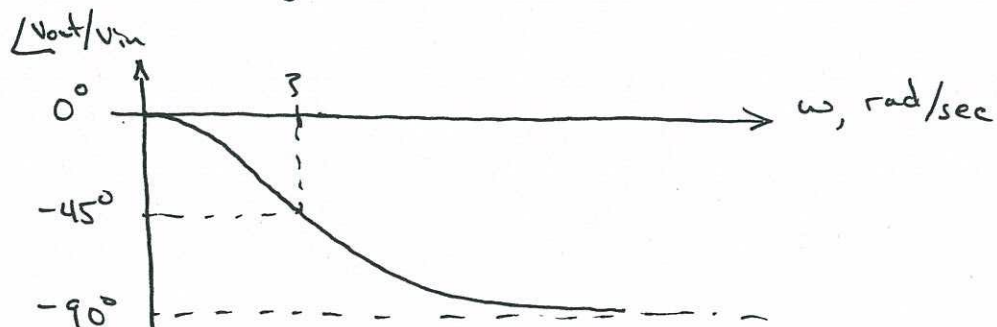
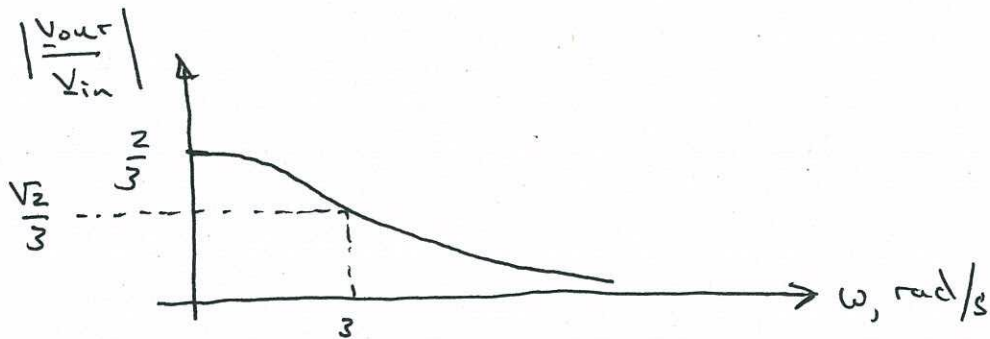
Frequency domain circuit:



$$V_{out} = V_{in} \left[\frac{\frac{2}{j\omega + 1}}{1 + \frac{2}{j\omega + 1}} \right] \Rightarrow \frac{V_{out}}{V_{in}} = \frac{2}{j\omega + 3}$$

$$\text{DC gain} = \frac{2}{j0 + 3} = \frac{2}{3}$$

$$\omega_c = 3 \text{ rad/sec}$$



11.2 (cont'd)

$$2 \frac{di}{dt} + 10i = 10v$$

Re-write in "standard" form:

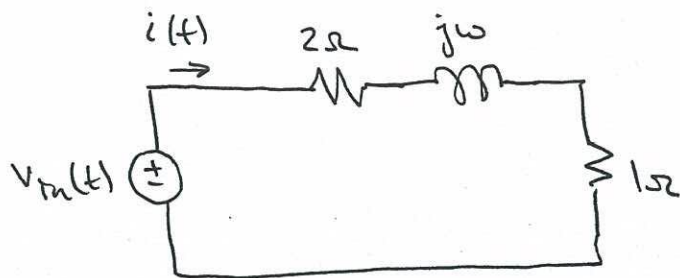
$$\frac{di}{dt} + 5i = 5v$$

↑

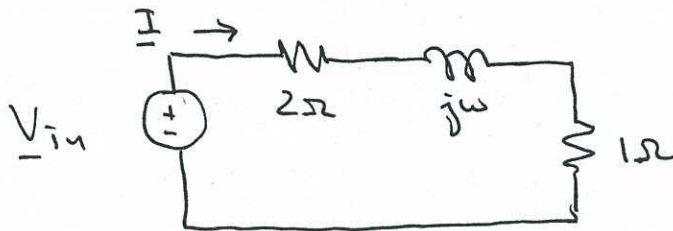
$$\frac{1}{\tau} = \omega_c$$

$$\underline{\underline{\omega_c = 5 \text{ rad/sec}}}$$

11.3



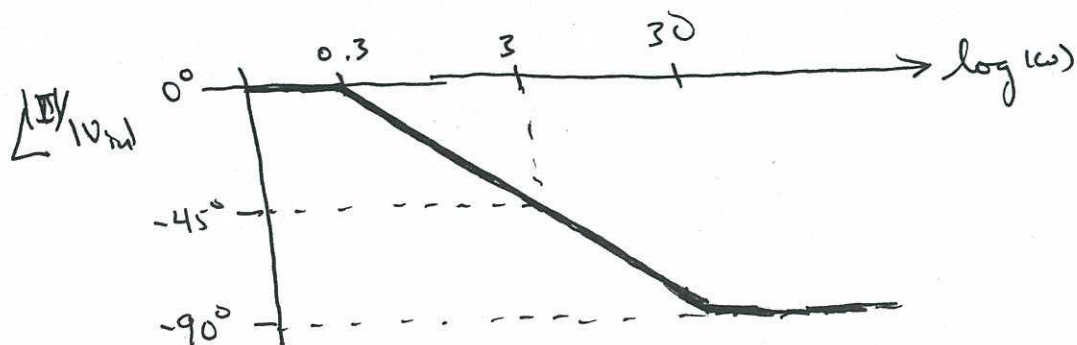
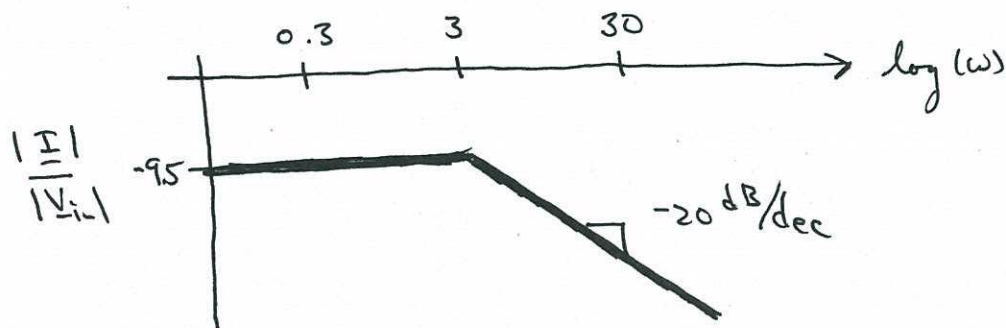
Frequency domain



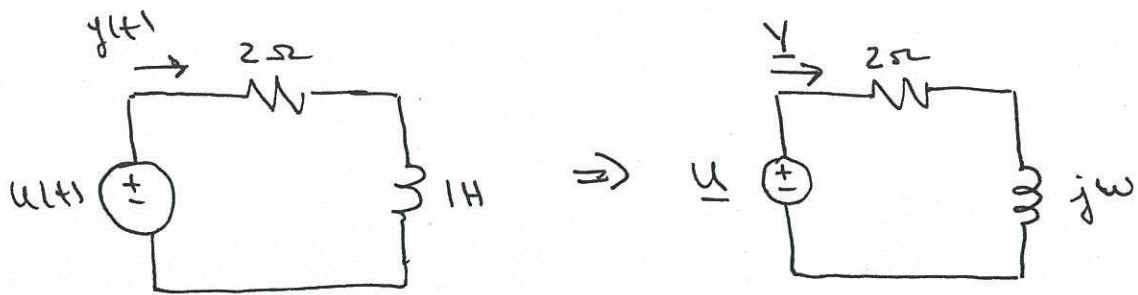
$$\underline{I} = \underline{V}_{in} \left[\frac{1}{j\omega + 3} \right] \Rightarrow \frac{\underline{I}}{\underline{V}_{in}} = \frac{1}{j\omega + 3}$$

$$\left| \frac{\underline{I}}{\underline{V}_{in}} \right| = \frac{1}{\sqrt{\omega^2 + 9}}, \quad \angle \frac{\underline{I}}{\underline{V}_{in}} = -\tan^{-1} \left(\frac{\omega}{3} \right)$$

$$\omega_c = 3 \text{ rad/sec}, \quad \text{DC gain} = \frac{1}{3} = -9.54 \text{ dB}$$



11.4

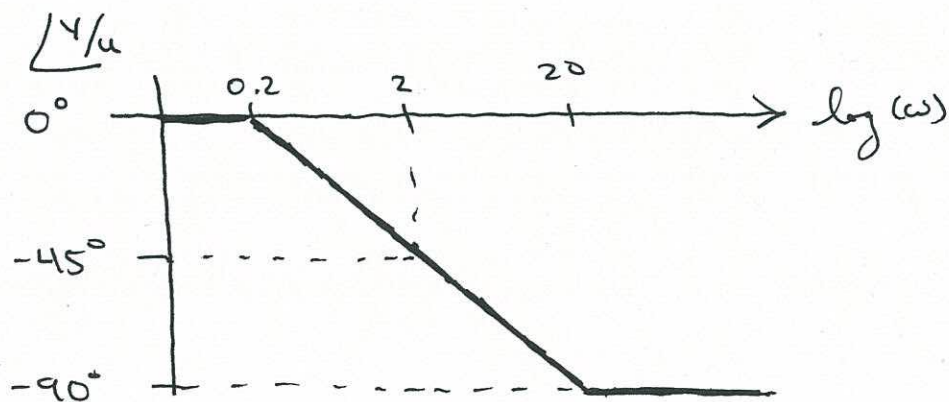
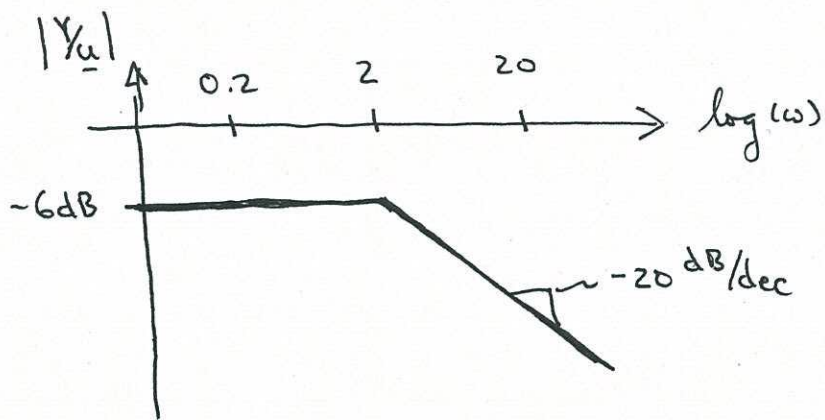


$$\underline{Y} = \frac{\underline{u}}{2 + j\omega} \Rightarrow \frac{\underline{Y}}{\underline{u}} = \frac{1}{2 + j\omega}$$

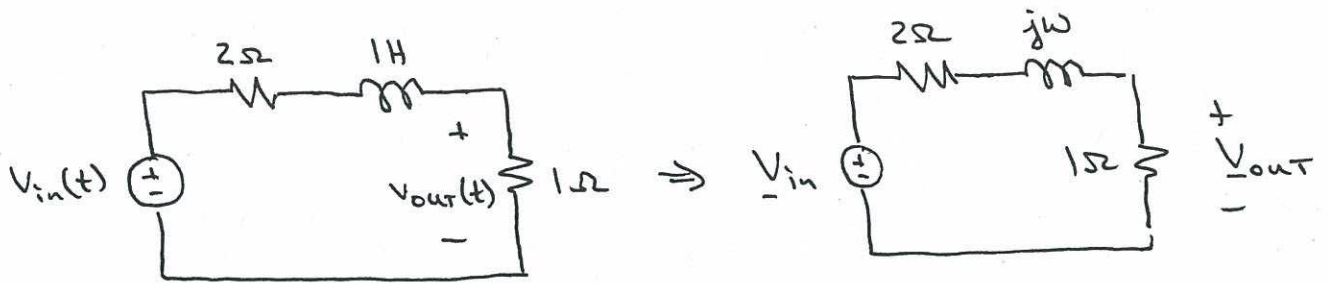
$$\left| \frac{\underline{Y}}{\underline{u}} \right| = \frac{1}{\sqrt{\omega^2 + 4}}, \quad \angle \frac{\underline{Y}}{\underline{u}} = -\tan^{-1}\left(\frac{\omega}{2}\right)$$

$$\text{DC gain} = \frac{1}{2} = -6 \text{ dB}$$

$$\omega_c = 2 \text{ rad/sec}$$



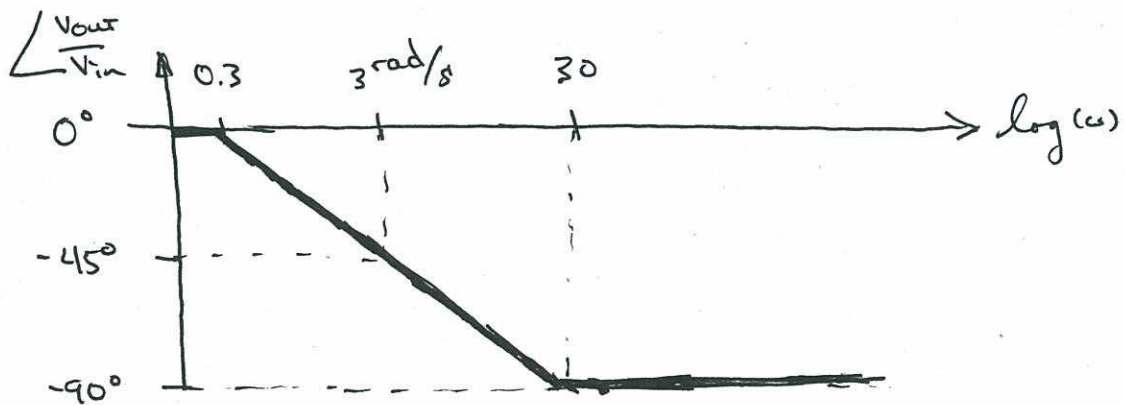
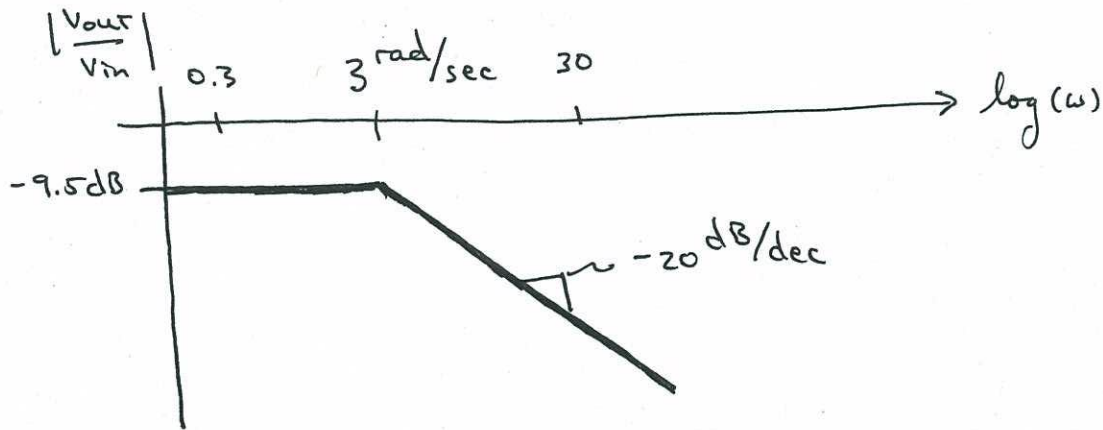
11.5



$$V_{out} = V_{in} \left[\frac{1\Omega}{2\Omega + j\omega + 1\Omega} \right] = \frac{1}{j\omega + 3} V_{in}$$

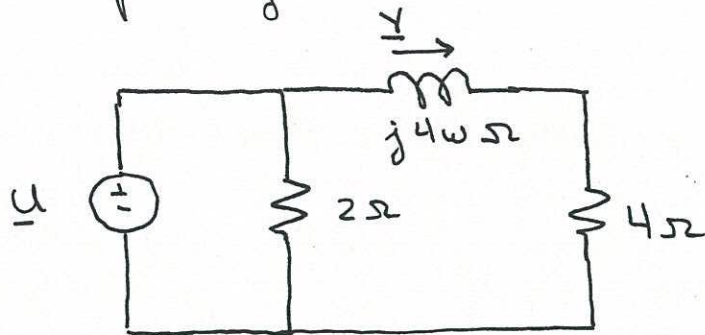
$$\text{DC gain} = \frac{1}{3} \Rightarrow \text{DC gain}_{dB} = 20 \log_{10}\left(\frac{1}{3}\right) = -9.54 \text{ dB}$$

$$\omega_c = 3 \text{ rad/sec}$$



11.6

Frequency domain circuit:

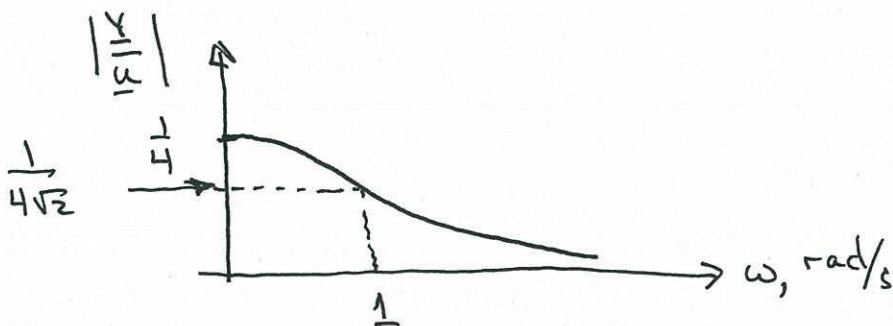


$$(a) \quad \underline{Y} = \underline{U} \left[\frac{1}{4 + j4\omega} \right] \Rightarrow \boxed{\frac{\underline{Y}}{\underline{U}} = \frac{1}{4 + j4\omega}}$$

$$(b) \quad \left| \frac{\underline{Y}}{\underline{U}} \right| = \left| \frac{1}{4 + j4\omega} \right| = \frac{1}{\sqrt{16 + 16\omega^2}} = \frac{1}{4\sqrt{1 + \omega^2}}$$

$$\text{DC gain} \Rightarrow \left| \frac{\underline{Y}}{\underline{U}} \right|_{\omega=0} = \frac{1}{4}$$

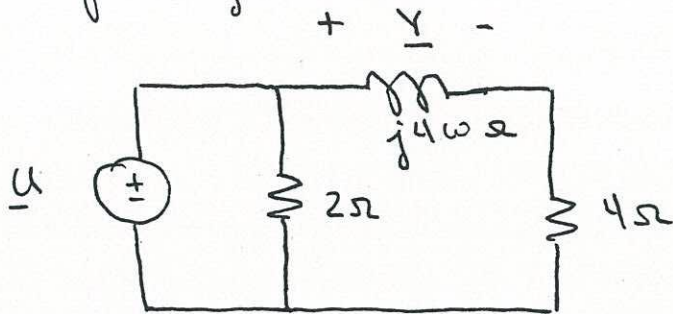
$$\omega_c = 1 \text{ rad/s}$$



(c) Low pass. Gain at low frequencies is non-zero. High frequencies are "stopped" by the circuit.

11.7

Frequency domain circuit:



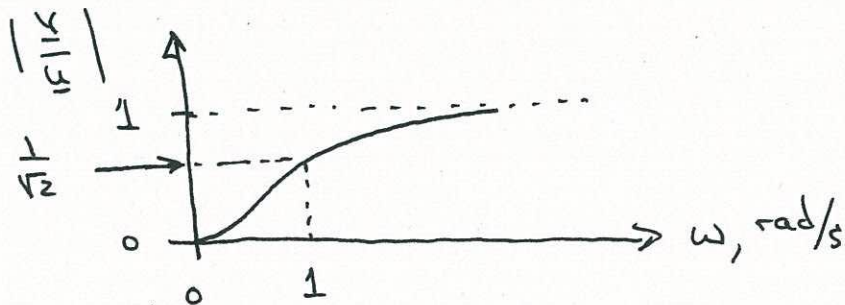
$$(a) \quad \underline{V} = \underline{U} \left[\frac{j4\omega}{4 + j4\omega} \right] \Rightarrow \boxed{\frac{\underline{V}}{\underline{U}} = \frac{j4\omega}{4 + j4\omega}}$$

$$(b) \quad \left| \frac{\underline{V}}{\underline{U}} \right| = \left| \frac{j4\omega}{4 + j4\omega} \right| = \left| \frac{j\omega}{1 + j\omega} \right|$$

$$\left| \frac{\underline{V}}{\underline{U}} \right| = \frac{\omega}{\sqrt{1 + \omega^2}}$$

$$\text{DC gain} = \left| \frac{\underline{V}}{\underline{U}} \right|_{\omega=0} = 0$$

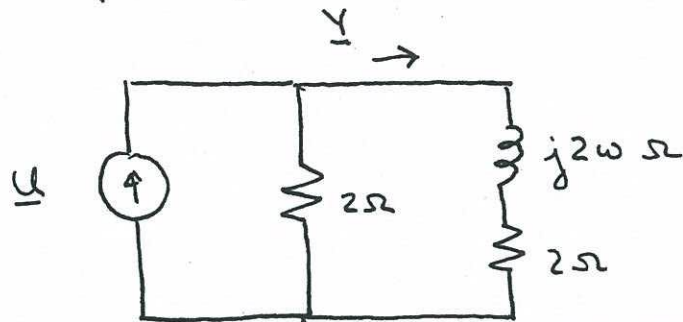
$$\omega_c = 1 \text{ rad/s}$$



(c) High-pass. Low frequencies are "stopped" by the filter.

11.8

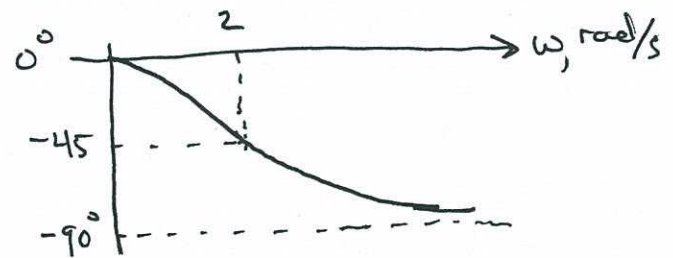
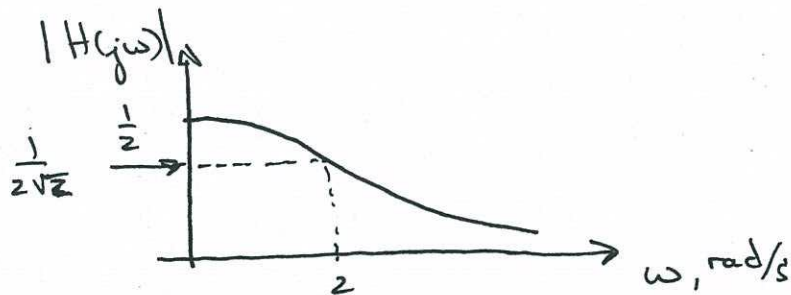
Frequency domain circuit:



$$(a) \quad \underline{Y} = \underline{U} \left[\frac{2}{4 + j2\omega} \right] \Rightarrow \boxed{H(j\omega) = \frac{\underline{Y}}{\underline{U}} = \frac{2}{4 + j2\omega}}$$

$$(b) \quad |H(j\omega)| = \frac{2}{\sqrt{16 + 4\omega^2}} = \frac{1}{\sqrt{4 + \omega^2}} \Rightarrow \begin{aligned} \text{DC gain} &= \frac{1}{2} \\ \omega_c &= 2 \text{ rad/sec} \end{aligned}$$

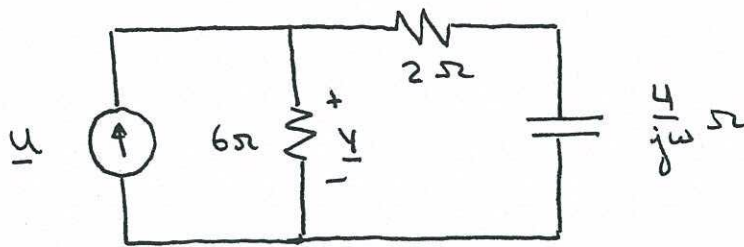
$$\angle H(j\omega) = 0^\circ - \tan^{-1}\left(\frac{\omega}{2}\right) \Rightarrow \begin{aligned} \angle H(j0) &= 0^\circ \\ \angle H(j\infty) &= -90^\circ \end{aligned}$$



(c) Low pass. High frequencies are "stopped" by the circuit.

11.9

Frequency domain circuit:

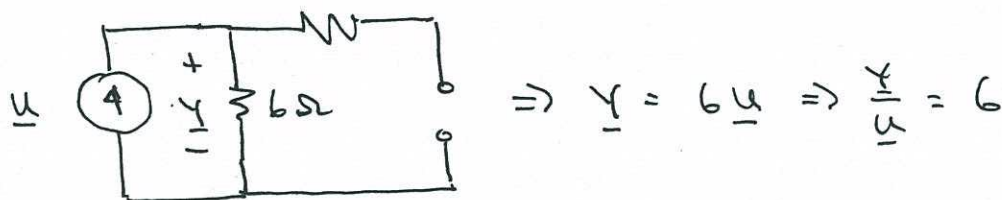


$$(a) \quad \underline{Y} = \underline{U} \underbrace{\left[\frac{2 + \frac{4}{j\omega}}{8 + \frac{4}{j\omega}} \right]}_{\substack{\text{current} \\ \text{divider}}} \times (6\Omega) \Rightarrow \boxed{\frac{\underline{Y}}{\underline{U}} = \frac{j12\omega + 24}{j8\omega + 4}}$$

current through
6Ω resistor.

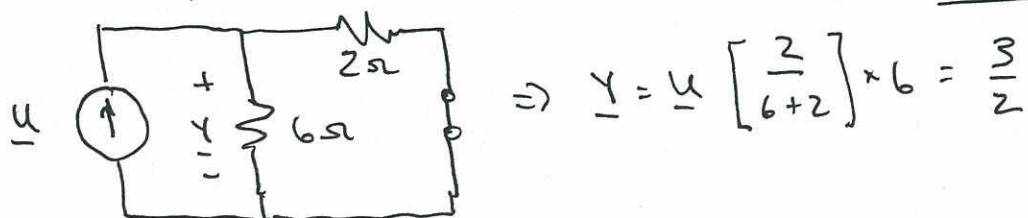
(b) Frequency response, $\omega \rightarrow 0$: $\frac{\underline{Y}}{\underline{U}} = \frac{24}{4} = 6$

circuit, $\omega \rightarrow 0$:



AGREE

Frequency response, $\omega \rightarrow \infty \Rightarrow \frac{\underline{Y}}{\underline{U}} \approx \frac{j12\omega}{j8\omega} = \frac{3}{2}$

circuit, $\omega \rightarrow \infty$ 

AGREE